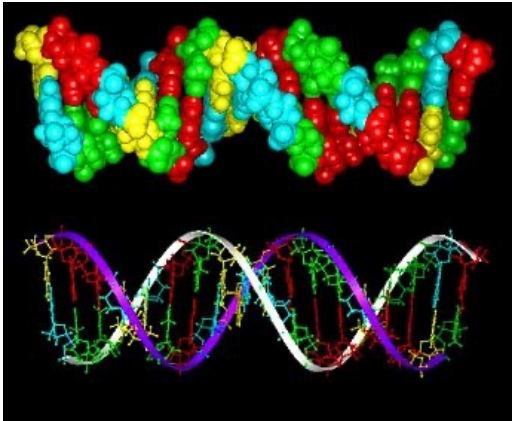


Polymers – Everywhere in Daily Life



DNA



Starch



Cellulose



Rubber



HDPE



Teflon



Nylon



PVC

Introduction to Polymers

- **What are polymers?**

- Polymers are organic (carbon-based) compounds where each molecule is a long-chain of covalently linked molecular units
- Think of it as a beaded necklace. Each bead is an unit that is repeated to form the necklace (polymer). There can be beads of one kind or of several different kinds.
- E.g. plastics, proteins, nucleic acids, carbohydrates

- **What is a monomer?**

- The basic chemical unit of a polymer, is called a monomer (beads in the necklace)
- E.g. vinyl chloride, ethylene glycol, lactic acid, amino acids, glucose, etc.

Some more terms

- **Macromolecules**

- Refers to very large molecules. Since polymers are generally composed of thousands of monomers, each polymer molecule is very large. Thus polymers are also termed as macromolecules

- **Oligomers**

- These are compounds having a few (unlike “many” in case of polymers) monomers joined together by covalent bonds (usually less than 50 repeating units). E.g. Oligonucleotides, peptides etc.

Types of Polymers: Chemical Structure

A-A-A-A-A-A-A-A-A-A-A-A-A-A

HOMOPOLYMERS

A-B-A-B-B-B-B-A-A-B-B-A-A-A-B

RANDOM COPOLYMERS

A-B-A-B-A-B-A-B-A-B-A-B-A-B-A

ALTERNATING COPOLYMERS

A-A-A-A-A-A-A-B-B-B-B-B-B-B-B

BLOCK COPOLYMERS

A-A-A-A-A-A-A-A-A-A-A-A-A-A-
|
B-B-B-B-B-B

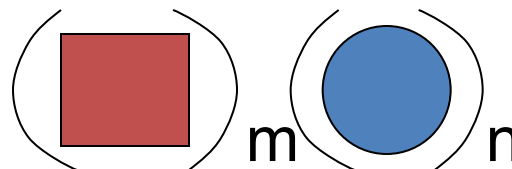
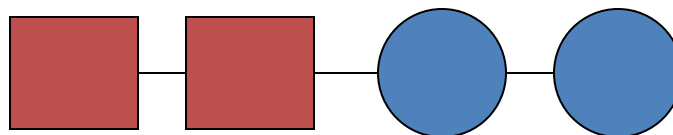
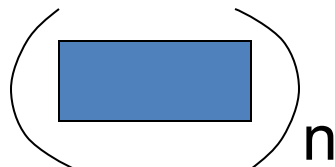
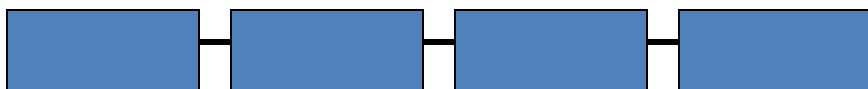
GRAFT COPOLYMERS

Polymer Nomenclature

Polymer types (classes) are named after the functional group linkage in their backbone i.e the bonds between the monomers:
E.g. Polyesters, Polyethers, Polyamides, Polyanhydrides etc.

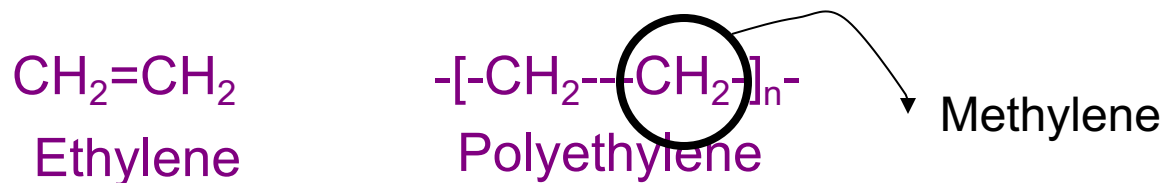
If more than one type of linkage is present then both figure in the names, e.g. polyetherimide

The language (script):



Polymer Nomenclature

- Most common naming process:
 - Poly(monomer), or
 - poly(monomer1-co-monomer2),
 - poly(monomer1-block-monomer2),
 - poly(monomer1-graft-monomer2) etc. i.e. source-based nomenclature
- IUPAC (International Union of Pure and Applied Chemistry) nomenclature: rarely used in common literature, only as references
- Example of a source-based polymer name:



IAPUC name of this polymer will be poly(methylene)

How are polymers synthesized?

Simply speaking: Individual monomer molecules (same or different kinds) are covalently linked together by a variety of chemical reactions to produce the polymer molecule.

- Carother's classification
 - **Addition polymerization**: Each repeat unit has the same number of atoms as the monomers
 - **Condensation polymerization**: Repeat units have less number of atoms than the monomers because of formation of small molecular weight reaction products (e.g. H_2O , HCl etc.)
- Modern classification: Based on how the reaction occurs
 - **Step polymerization**
 - random union of monomer molecules
 - **Chain polymerization**
 - polymer chain grows only by reaction of a monomer with a reactive end-group on the growing chain

Molecular weight (molar mass)

Molar mass of a polymer = Mass of one mole of the polymer (i.e. 6.023×10^{23} molecules of the polymer). Usually expressed in units of g/mol or daltons. Molecular weight of 100,000 daltons means molar mass of 100,000 g/mol i.e. one mole weighs 100,000 grams

For a homopolymer: molar mass (M) = xM_o , where M_o is the molar mass of a repeat unit (monomer) and x = average degree of polymerization (i.e. the number of repeat units in a polymer chain)

The polymerization process inherently creates polymer molecules of different lengths. Therefore **we always measure “AVERAGE” molecular weights and “AVERAGE” degree of polymerization for polymers.**

Average molecular weights

Number average molecular weight (M_n):

Defined as $M_n = \sum X_i M_i$, where X_i = mole fraction of molecules having a molar mass of M_i

If N_i = Number of moles with mass M_i , then

$$M_n = \sum N_i M_i / \sum N_i \dots\dots\dots(1)$$

NOTE: This is the arithmetic mean of the molar mass distribution

$N_i M_i$ is also same as w_i , the weight of molecules having a molar mass of M_i i.e. $M_n = \sum w_i / \sum (w_i / M_i)$

Example 1: Polymer sample consisting of 9 moles with molecular weight 30,000 and 5 moles with molecular weight 50,000.

$$\begin{aligned} M_n &= \{[9 \text{ mol} \times 30000 \text{ g/mol}] + [5 \text{ mol} \times 50000 \text{ g/mol}]\} / (9+5) \text{ mol} \\ &= 37,000 \text{ g/mol} \end{aligned}$$

Example 2: Sample with 9 grams of molecular weight 30,000 and 5 grams with molecular weight 50,000

$$M_n = (9 + 5) \text{ g} / \{[9 \text{ g} / 30,000 \text{ g/mol}] + [5 \text{ g} / 50000 \text{ g/mol}]\} = 35,000 \text{ g/mol}$$

Average molecular weights

Weight average molecular weight (M_w):

Defined as $M_w = \sum W_i M_i$, where W_i = weight fraction of molecules having a molar mass of M_i

If N_i = Number of moles with mass M_i , then

$$W_i = N_i M_i / \sum N_i M_i = w_i / \sum w_i \dots\dots\dots(2)$$

$$\text{Therefore, } M_w = \sum N_i M_i^2 / \sum N_i M_i = \sum w_i M_i / \sum w_i \dots\dots\dots(3)$$

[Second order average]

Example 1: Polymer sample consisting of 9 moles with molecular weight 30,000 and 5 moles with molecular weight 50,000.

$$\begin{aligned} M_w &= \{9 \text{ mol } (30000 \text{ g/mol})^2 + 5 \text{ mol } (50000 \text{ g/mol})^2\} \text{ divided by} \\ &\quad \{9 \text{ mol } (30,000 \text{ g/mol}) + 5 \text{ mol } (50,000 \text{ g/mol})\} \\ &= 40,000 \text{ g/mol} \end{aligned}$$

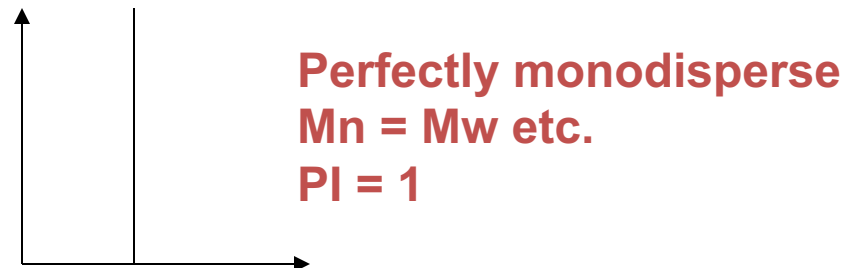
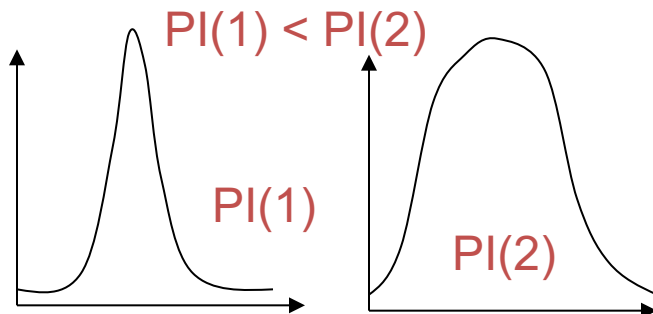
Example 2: Sample with 9 grams of molecular weight 30,000 and 5 grams with molecular weight 50,000

$$M_w = \{9\text{g } (30,000 \text{ g/mol}) + 5\text{g } (50000 \text{ g/mol})\} / (9 + 5) \text{ g} = 37,000 \text{ g/mol}$$

NOTE: M_w IS ALWAYS GREATER THAN or EQUAL TO M_n

Polydispersity index (PI)

- Polydispersity index is a measure of the width of molecular weight distribution
- It is defined as the ratio between the weight and number average molecular weights (M_w / M_n)
- Therefore, by definition **PI is ≥ 1 , since $M_w \geq M_n$**
- For a perfectly monodisperse polymer $PI = 1$
- Greater the PI, broader is the molecular weight distribution

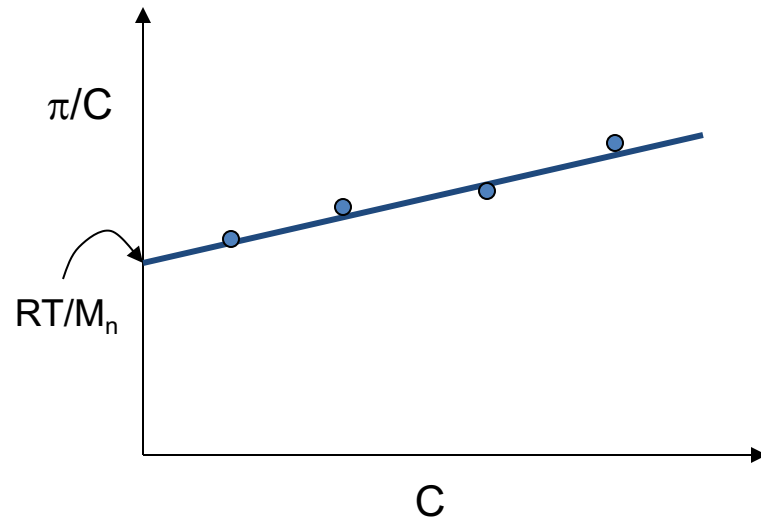
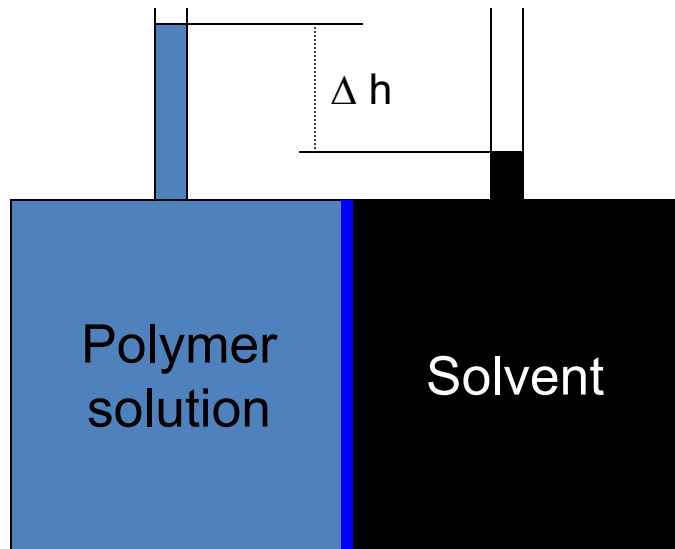


Molecular weight measurements

- **Number average molecular weight** (Calculated with techniques that measure “number” of particles – colligative properties)
 - End group analysis
 - Membrane osmometry
 - Vapor pressure osmometry
 - Cryoscopy
 - Refractive index measurements
- **Weight average molecular weight**
 - Light scattering
 - Ultra-centrifugation
- **Viscosity measurements**
- **Molecular weight distribution:**
 - Gel permeation Chromatography
 - Mass spectrometry
 - Fractional solution
 - Thin layer chromatography

Membrane osmometry

- Widely used method for measuring M_n (measuring colligative property)
- Ideal for polymers with M_n between 50,000 and 2 million



Osmotic pressure is related to molecular weight by the van't Hoff equation

$$(\pi/C) = (RT/M_n) + A_2C \dots\dots\dots(1)$$

Where π , the osmotic pressure is given by $\pi = \rho g \Delta h \dots(2)$

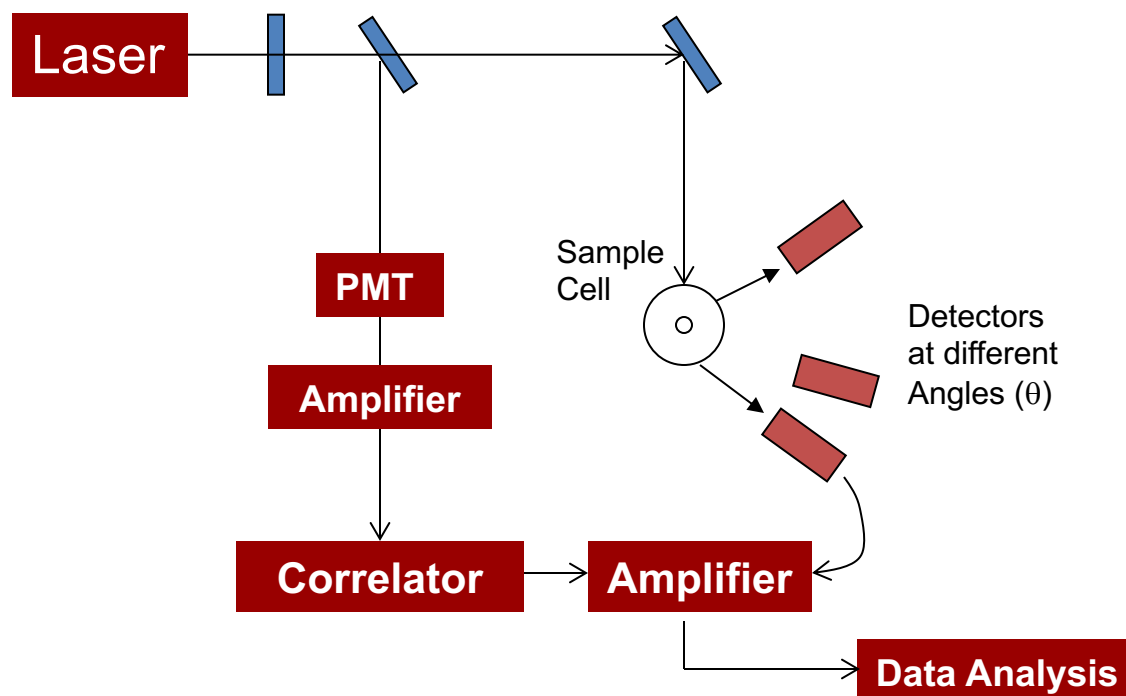
R = Gas constant, T = Temp ($^{\circ}K$), C = conc. in g/L, ρ = solvent density (g/cc)

g = acceleration due to gravity (m/s^2) and Δh = as shown in figure

At $C = 0$ (extrapolation), the intercept provides M_n

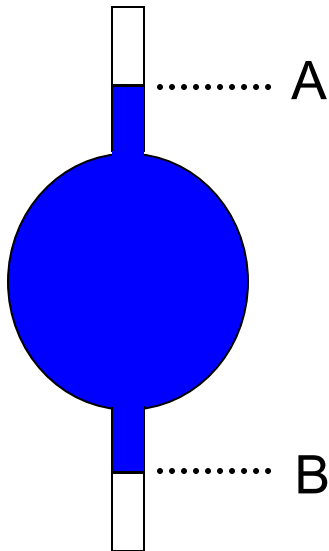
Light scattering: Setup

- Widely used for molecular weight determination
- Gives Mw
- Also used to measure hydrodynamic radius
- Useful between molecular weights of 10,000 to 10 million



Viscosity Measurements

- Widely used method to calculate **molar mass M_n (viscosity average)**
- Lot of manufacturer's will only specify **intrinsic viscosity** values in their CA, instead of molecular weights
- **Higher the molar mass, higher is the viscosity**



Efflux time: Time for the polymer solution to travel from A to B (measured with a stop watch)

More time → more viscous → higher molecular weight

Viscosity Measurements

Efflux time of pure solvent (no polymer) = t_o

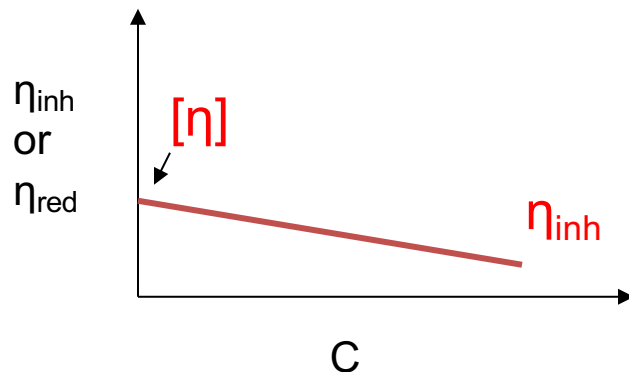
Efflux time of polymer solution at concentration “c” = t_c

Relative viscosity = t_c / t_o (always greater than 1) = η_r

Inherent viscosity = $\ln(\eta_r) / c = \eta_{inh}$

Intrinsic viscosity $[\eta]$ is defined as

$$[\eta] = \lim_{c \rightarrow 0} (\eta_{inh})$$



Molecular weight and viscosity is related by the Mark-Houwink-Sakurada equation: $[\eta] = KM^a$

K and a are experimentally determined from $\log [\eta]$ vs $\log [M]$ plots

Gel permeation chromatography (GPC) or Size exclusion chromatography (SEC)

- Extremely powerful method for determining molecular weight distribution of a polymer.
- Specialized columns packed with beads of a porous gel with porosity in the range of $50 - 10^6 \text{ \AA}^3$

Morphological properties:

Crystallinity

- When a polymer is slowly cooled from the melt state, it can form ordered structures (similar to crystals). Such polymers are called crystalline (semi-crystalline)
- Polymers with regular, compact structures and strong intermolecular forces, such as hydrogen bonds have high degrees of crystallinity
 - as crystallinity increases, the polymer becomes more opaque due to scattering of light by the crystalline regions
- **Melt transition temperature, T_m** (or Crystalline melting point): The temperature at which all crystalline regions melt i.e. crystallinity disappears
 - as the degree of crystallinity increases, T_m increases

Morphological properties: Crystallinity

Crystallinity has major effects on

a. Mechanical properties

Increased stiffness

Less flexible

b. Diffusion rates

Less permeable to diffusing molecules

c. Rates of hydrolysis

Low penetration of water and hence slow hydrolysis

Morphological properties:

Amorphousness

- Polymers that doesn't form an ordered structure when cooled from a melt are called amorphous polymers
- Amorphous polymers are also referred to as glassy polymers
 - They lack crystalline domains that scatter light and are **transparent**
 - On heating, amorphous polymers are transformed from a hard glass (stiff, brittle) to a soft, flexible, rubbery state
- **Glass transition temperature, T_g :**
 - The temperature at which a polymer undergoes a transition from a hard glassy state (amorphous solid) to a rubbery state

$$T_g < T_m$$

Glass Transition Temperature (T_g)

- **Motion of polymer chains** possible above T_g , while below T_g only atomic or small molecule motions occur
- Number of **polymer properties e.g. heat capacity, density, permeability, dielectric constant etc. changes** abruptly at T_g
- Polymers are **brittle below T_g and flexible (rubbery) above T_g . Rubber ball shatters in liquid nitrogen.**
- **Depending on usage, proper polymer is chosen**
 - Silicones have an extremely low T_g and T_m , they are always fluid at relevant temperatures (breast implants)
 - Polystyrene, PMMA is in a glassy state at room temp and above ($T_g \sim 100^\circ\text{C}$). So they can be made into hard plastic objects
- Chain mobility is critical in diffusion. **Diffusion below T_g is much lower than diffusion above T_g**

Morphology: Effect of cooling rate

- **Poly(ethylene terephthalate)**, abbreviated PET can be made with crystalline domains ranging from 0% to 55% depending on how it is processed. Completely amorphous PET is formed by **cooling the melt quickly**
 - plastic beverage bottles are PET: What do u think they are?
 - with a low degree of crystallinity
- By **cooling slowly**, more molecular diffusion occurs, chains become more ordered and crystalline domains form
 - PET with a high degree of crystallinity can be drawn into textile fibers and tire cords (Dacron)

Factors affecting T_g

- **Molecular weight:** End groups have higher mobility (higher free volume). Low molecular weight polymers have higher number of ends per unit volume and hence lower T_g . Higher molecular weight polymers have higher entanglement, less mobility and hence higher T_g .
- **Bulky side groups** hinders chain motion making chains stiffer. Hence higher T_g (PMA vs. PMMA)
- **Strong intermolecular attractions** $\rightarrow$ higher T_g

Higher $T_m \rightarrow$ Higher T_g

Can be measured using Differential Scanning Calorimetry

Biomedical Polymers

- **Biopolymers:** Polymers that can be safely used in biological or medical applications or polymers that are naturally present in organisms are called biopolymers
- **Synthetic biopolymers**
 - Chemically synthesized polymers (does not occur in nature)
 - Designed specifically for particular usage, including bio-molecule delivery, implants, tissue engineering, wound dressing, prosthetics etc.
- **Natural biopolymers**
 - Occurs naturally (in plants and other organisms)
 - Isolated and purified for biomedical usage including drug delivery, tissue engineering, wound applications etc.

Advances in Biomaterials for Drug Delivery

Owen S. Fenton, Katy N. Olafson, Padmini S. Pillai, Michael J. Mitchell,
and Robert Langer**

Advances in biomaterials for drug delivery are enabling significant progress in biology and medicine. Multidisciplinary collaborations between physical scientists, engineers, biologists, and clinicians generate innovative strategies and materials to treat a range of diseases. Specifically, recent advances include major breakthroughs in materials for cancer immunotherapy, autoimmune diseases, and genome editing. Here, strategies for the design and implementation of biomaterials for drug delivery are reviewed. A brief history of the

lose their pharmacological activity due to changes in environmental factors such as moisture, temperature, and pH, which can occur in the body or during storage. As the biotechnology industry continues to develop new classes of biopharmaceuticals, improved fundamental understanding of how drug delivery affects safety and efficacy, along with new delivery

Synthetic vs. natural biopolymers

Synthetic polymers	Natural polymers
Chemically synthesized from monomers <i>E.g: PLGA, PET, PEG etc.</i>	Isolated as polymers from plants or organisms <i>E.g.: Cellulose, chitosan, hyaluronic acid, chondroitin sulfate, gelatin, collagen, proteins and DNA etc.</i>
Can be tailored according to desired properties	In its native form, the properties cannot be controlled
Can be modified or functionalized according to applications	Modification is often difficult, but some polymers can be modified to alter properties for specific usage (semi-synthetic)
Large scale production is easily feasible	Large scale production can be often difficult

Natural polymers in biomedical applications

Table 1. A summary of the main properties and applications of polymeric biomaterials

Polymer	Main applications and comments
Natural polymers	
Proteins and protein-based polymers	Absorbable, biocompatible, nontoxic, naturally available, typically elastic materials used as implants and in tissue engineering.
Collagen	Absorbable sutures, sponge wound dressing, drug delivery microspheres.
Albumin	Used in cell and drug microencapsulation.
Poly(amino acids)	Usually poly(α ,L-amino acids), examples include poly(α ,L-lysine), poly(α ,L-glutamic acid), poly(aspartic acid) etc. Advantages: nontoxic, nonantigenic and biocompatible. Used as oligomeric drug carriers.
Polysaccharides and derivatives	
<i>From vegetable sources</i>	
Carboxymethyl cellulose	Cell immobilization via a combination of ionotropic gelation and polyelectrolyte complex formation (e.g. with chitosan), in drug-delivery systems and dialysis membranes.
Cellulose sulphate	Component of polyelectrolyte complexes for immunoisolation. Complex-forming ability is highly sensitive to degree of acylation.
Agarose	Largely used as supporting materials in clinical analysis and as an immobilization matrix.
Alginate (marine sources, algae)	Excellent gel-formation properties; its relative biocompatibility; microstructure and viscosity are dependent on the chemical composition (batch-to-batch variations). Used as immobilization matrices for cells and enzymes, controlled release of bioactive substances, injectable microcapsules for treating neurodegenerative and hormone-deficiency diseases.
Carrageenan	Excellent thermoreversible properties. Used for microencapsulation.
<i>From human and animal sources</i>	
Hyaluronic acid	Excellent lubricant, potential therapeutic agent.
Heparin and heparin-like glycosaminoglycans	Antithrombotic and anticoagulant properties. Extensively used in surgery. Some are candidates for ionotropic gelation and capsule formation.
<i>Microbial polysaccharides</i>	
Dextran and its derivatives	Excellent rheological properties. Plasma expander. Widely used as a drug carrier.
Chitosan and its derivatives	Biocompatible, nontoxic, excellent gel- and film-forming ability, natural polycation. Widely used in controlled-delivery systems (e.g. gels, membranes, microspheres).

Desirable properties of biopolymers

- Non-immunogenic (Low inflammatory response)
- Non toxic (does not cause cell or tissue death / damage)
- Properties (mechanical, chemical, electrical) can be tailored for specific applications
- Properties are conducive of easy scale-up (i.e. mass production)

In some cases (especially in most drug delivery applications) it is desirable that the polymer does not remain in the body for a prolonged period. It should degrade and adsorbed, excreted or metabolized under physiological conditions.

Note: Such physiologic degradation or erosion is NOT desirable for a large number of biomedical applications (implants, prosthetics etc.)

Choosing biomedical polymers

- Application.... Application..... Application?
- Route of administration
 - Intravenous, parenteral, cutaneous, mucosal
- Bio-compatibility
 - What kind of cells and tissues will be exposed, host interactions
- Is bio-erosion important or even required?
 - Permanent or temporary material
- Surface properties and compositions
 - Is protein adhesion necessary or deleterious, interactions with other cells, specific or non-specific targeting

Choosing biomedical polymers (cont.)

- Mechanical properties
 - How much load does the device need to bear, how long?
 - Does one need to produce arbitrary shapes and designs
- Environmentally sensitive
 - Stimuli sensitive properties, swelling or drug release?
- Permeability
 - Solute diffusion, porosity

Box 1. Polymer properties needed for specific biomaterial applications

Dental applications

Properties and design requirements

Stability and corrosion resistance in the aggressive oral environment; plasticity and easy processing; strength and fatigue resistance; coating activity; good adhesion and integration to the surrounding tissues; low allergenicity; long lifetime.

Applications and polymers used to date

Poly(methylmethacrylate)-based resins as filling materials; prosthesis. Mechanical properties improved by copolymerization. Poly(zinc acrylates); polyamides.

Future needs

New materials with improved elasticity and bone-soft-tissue integration.

Ocular

Properties and design requirements

Composition and design: solutions, ointments, gels and films. Adhesion, gel- or film forming ability, solubility or hydrophilicity. Biocompatibility and site-specific bioadhesive properties.

Applications and polymers used to date

Hydroxypropylmethylcellulose and poly(vinyl alcohol) for artificial tears; hyaluronic acid; polyacrylamide gels.

Future needs

Polymeric biomaterials with 'intelligent' surfaces to improve bioadhesion.

Orthopaedic

Properties and design requirements

Strength and resistance to mechanical constraints and fatigue; stress modelling; surface designed for cell-specific interactions; sterilizability; good bio- and histocompatibility; durability (for long-term bone implants); good integration with bone and muscles; easily fabricated; injectable or *in situ* moulding, depending on use.

Applications and polymers used to date

Bone cements; degradable scaffold for bone and cartilage replacement; cell and growth-factor delivery. Poly(methylmethacrylate); poly(lactic acid), poly(glycolic acid) and their copolymers.

Future needs

New synthetic biodegradable scaffolds with improved biological and mechanical properties.

Vascular devices

Properties and design requirements

Wear-and-tear resistance; fatigue resistance; lubricity; lack of thrombus or embolism formation and chronic inflammatory response; biostability; sterilizability; porosity (tissue ingrowth); mechanical integrity.

Applications and polymers used to date

Catheters, vascular graft materials, heart valves. Silicones, Teflon®, poly(urethanes).

Future needs

Interdisciplinary approach to create biomimetic materials with enhanced cell interaction; bioresponsive polymers; computer-aided design of micro- and ultrastructures of the implants.

Biortificial organs and soft-tissue implants

Properties and design requirements

The same mechanical properties as the surrounding tissues; no chronic response to toxic leachables; integration with natural tissues without fibrous reaction; porosity; creep resistance.

Applications and polymers used to date

Artificial skin and breast implants (collagen and silicone based). Artificial liver (hepatocyte transplantation) – poly(lactic acid) and poly(glycolic acid).

Future needs

Creation of complex, three-dimensional structures that can be used like artificial organs. Development of a total artificial heart.

Encapsulation

Properties and design requirements

Permeability and diffusive properties to allow optimum transport through the membrane; membrane thickness and mechanical stability must remain intact in physiological environments.

Applications and polymers used to date

Hydrogel- and polyelectrolyte-complex-based microcapsules for encapsulating cells, enzymes, hormones, proteins for bioartificial organs and disease treatments. Alginate, chitosan, synthetic polyelectrolytes.

Future needs

New polymers and strategies for encapsulation can result in the preparation of more-effective encapsulation systems; responsive polymers, novel techniques, non-traumatic implantation procedures to preserve the integrity of the protective membrane.

Drug delivery

Properties and design requirements

Design depends on the route of administration and release site; well-defined structure and functionality.

Applications and polymers used to date

Hydrogels, membranes, networks and copolymers with different structures and applications.

Future needs

Improving the design of the systems, new bioresponsive and 'intelligent' materials.

Biocompatibility and Biodegradability

- Biocompatibility

- Property of materials that determines the extent of adverse physiological reactions (inflammation, immune response, toxicity etc.) when the material is placed inside the body. Lower the adverse reactions, more biocompatible is the material.

- Biodegradability

- In general refers to the ability of materials (polymers) to break down into smaller units (by physiological conditions) that are either adsorbed (metabolized) or excreted by the body.
- This is a very generalized term used interchangeably with other similar terms: bio-erosion, bio-resorption etc. We will visit this issue in more details later in this course

Biocompatibility: Host reactions to polymers

- Mostly a result of normal physiological processes that are designed by nature to act as host defense. The key is to use polymers that the body does not get “mad” about i.e. can tolerate and co-exist
- All materials (polymers) will interact with the host. It is the extent of that interaction that is important
- Some key interactions:
 - Blood-material interactions
 - Protein adsorption, coagulation, platelet adhesion
 - Complement activation
 - Leukocyte adhesion / activation
 - Encapsulation , scar tissue formation
 - Foreign body reaction
 - Infection and inflammation

All these will be covered in the inflammation part of the course

Assessment of Biocompatibility

- **Cyto-toxicity measurements**
 - Morphological cell culture assays
 - Direct contact
 - Elution or extract dilution
 - Biochemical cell function assays
 - Enzymatic activity following polymer exposure
- **In-vivo tissue reactions**
 - Histological evaluation following implantation
 - Time point studies to determine extent of reactions
- **Blood compatibility of polymers**
 - Coagulation studies
 - Hemolysis measurements
 - In-vivo experiments to evaluate systemic toxicity

Improving biocompatibility

- Surface modification with more compatible polymers. E.g. Polyethylene glycol, hyaluronic acid (biomimetic polymers)
- Surface modification to reduce protein adsorption (i.e. reduce blood interactions) – e.g. making the surface more hydrophilic, less cationic etc.
- Concurrent release or delivery of anti-inflammatory agents at the implant site
- Using alternative routes to avoid systemic toxicity (delivery into the blood would have higher chances of systemic toxicity)

Biodegradability

- Most drug delivery devices are by necessity temporary → widest use of biodegradable polymers
- Lets take care of the terminology first (Consensus Conference of The European Society of Biomaterials, 1987)
 - **Biodegradation**: Degradation by biological molecules e.g. enzymatic or microbial degradation.
 - **Bioerosion**: Erosion of a polymer into water soluble products under physiological conditions. This includes both physical and chemical processes (suggested by Heller, 1987)

Physical modes of bioerosion

- **Bulk erosion**

- The rate of water penetration into the solid device exceeds the rate at which the polymer is eroded (i.e. transformed into water soluble products)
 - leads to an erosion process that occurs throughout the entire volume of the solid device (particle or implant)
 - most hydrophilic polymers, devices with high porosities would be bulk eroding (this is the case with most polymers currently used in drug delivery)

- **Surface erosion**

- The rate at which water penetrates into the polymeric device is slower than the rate of erosion
 - The polymer starts eroding before water has penetrated the entire volume of the device
 - leads to erosion on the device (particle or implants) surface
 - device will become thinner or smaller over time.
 - Hydrolytically surface eroding: Polyanhydrides. Poly(orthoesters). Note this is also dependent on fabrication process and device geometry.

Factors influencing hydrolytic bio-erosion

- **Backbone hydrolysis is the most common mechanism of erosion** for currently available synthetic biopolymers.
- Main **factors determining overall rate of erosion**:
 - **Chemical stability of backbone: How labile are the linkages?**
 - Susceptibility to hydrolytic cleavage: Anhydrides > Esters > Amides → Faster erosion
 - **Hydrophobicity of the monomer: ability of water molecules to penetrate into the polymer structure.**
 - E.g. Erosion of poly(anhydrides) slows down ~ three orders of magnitude by replacing the hydrophilic monomer sebacic acid with suitable hydrophobic monomers
 - **Morphology of the polymer:** Crystalline polymers are dense and resist water penetration. Amorphous polymers with $T_g > 37^\circ\text{C}$ are at glassy state while $T_g < 37^\circ\text{C}$ are at rubber state. Glassy polymers are less permeable to water.
 - **Molecular weight, fabrication process** etc
 - **Geometry of the implanted device:** Surface to volume ratio

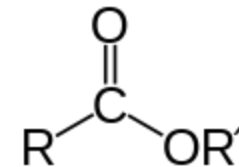
Host-induced hydrolytic processes

- **Simple hydrolysis:** Action of neutral water. E.g. polyglycolic acid erosion.
- **Ion-catalyzed hydrolysis:** Ions present in extracellular fluids (PO_4^{-3} , H^+ , OH^- , Na^+ , Cl^- , Mg^{2+} , Ca^{2+} etc.). Certain ions can catalyze hydrolytic reactions. E.g. polyester hydrolysis rate increases several folds in presence of cations
- Local **pH changes** due to inflammatory reactions have been shown to accelerate hydrolysis
- **Proteases, elastases, and other enzymes** increase the rates of hydrolysis, especially for polyesters and polyamides

Oxidative biodegradation

- Free radical initiated oxidation of polymers → significant for polyethers (PEG), polyamides etc.
 - Host induced: Activated phagocytic cells (macrophages, neutrophils etc.) produce directly or indirectly, superoxide anion, hydrogen peroxide, all of which are powerful oxidants.
 - Environment mediated: Metal ion induced oxidation (polyether urethane). Cracks on the polymer device.

Biodegradable polymers: Polyesters



- Polymers with **ester linkages between monomers**
- **Fast hydrolysis of ester bond** → **bulk degradation** with random hydrolytic scission of the polymer backbone
- Most common synthesis process: **Ring opening polymerization of the corresponding cyclic monomer** → **Catalysts (most commonly stannous octoate) are routinely used** to increase polymerization efficiency and yield high molecular weight polymers

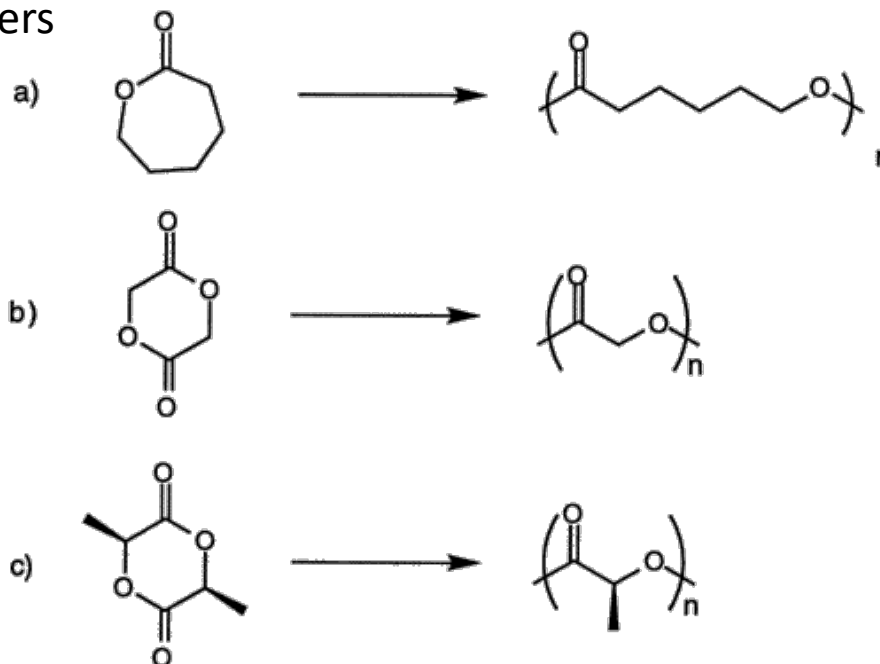
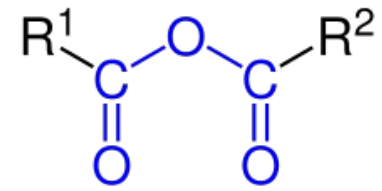


Figure: Ring-opening polymerization of selected cyclic lactones to give the following: (a) poly(-caprolactone) (PCL); (b) poly(glycolic acid) (PGA); (c) poly(L-lactic acid) (PLA).

Polyesters: PLA, PGA, PLGA

- Most extensively used biodegradable polymers
 - FDA approved as bio-absorbable sutures
 - Used extensively as particles in drug and gene delivery
 - Tissue engineering scaffolds
- Hydrolysable into lactic acid and glycolic acid
 - Lactic and glycolic acid are normal physiological products
 - Ultimately metabolized by the body into CO_2 and H_2O
 - However, **accumulation of acids creates local pH drop causing irritation**
 - Hydrolysis is catalyzed by acids and bases
 - PLA hydrolysis is slower than PGA (bulky methyl group), PLGA hydrolysis depends upon composition
 - Since Poly(L-lactic acid) is highly crystalline compared to Poly(DL-lactic acid) or PLGA it has much higher T_g and slower rate of hydrolysis.
 - Polymers with acid end groups ($-\text{COOH}$) hydrolyses faster $\rightarrow$ acid end group catalysis

Polyanhydrides



- Surface erosion can be achieved by synthesizing polymers that are hydrophobic (prevents water from entering device) yet contains highly water-sensitive linkages → polyanhydrides
- **Produces near zero-order release kinetics**
 - Constant rate of release due to surface degradation only
- **Degradation rate can be controlled by polymer composition**
 - Changing the hydrophobic monomers → aliphatic anhydrides degrade in days whereas aromatic ones can take years
- **Biocompatible**
 - Has been used in clinical trials for brain cancer patients (drug release implants)

Structures of representative polyanhydrides and their copolymers used in localized delivery applications

Polymer	Structure
<i>Aliphatic polyanhydrides</i>	
PSA	$\left[\text{C}(=\text{O})-(\text{CH}_2)_8-\text{C}(=\text{O})-\text{O} \right]_n$
<i>Aromatic polyanhydrides</i>	
P(CPV)	$\left[\text{C}(=\text{O})-\text{C}_6\text{H}_4-\text{O}-(\text{CH}_2)_4-\text{C}(=\text{O})-\text{O} \right]_n$
P(TA-IPA)	$\left[\text{O}-\text{C}(=\text{O})-\text{C}_6\text{H}_4-\text{CH}_2-\text{O}-\text{C}(=\text{O})-\text{C}_6\text{H}_4-\text{C}(=\text{O})-\text{O} \right]_n$
P(CPP-IPA)	$\left[\text{C}(=\text{O})-\text{C}_6\text{H}_4-\text{O}-\text{CH}_2\text{CH}_2\text{CH}_2-\text{O}-\text{C}(=\text{O})-\text{C}_6\text{H}_4-\text{C}(=\text{O})-\text{O}-\text{C}(=\text{O})-\text{C}_6\text{H}_4-\text{C}(=\text{O})-\text{O} \right]_n$
<i>Aliphatic-Aromatic Polyanhydrides</i>	
P(CPP-SA)	$\left[\text{C}(=\text{O})-\text{C}_6\text{H}_4-(\text{CH}_2)_3-\text{C}_6\text{H}_4-\text{C}(=\text{O})-\text{O} \right]_n \left[\text{C}(=\text{O})-(\text{CH}_2)_8-\text{C}(=\text{O})-\text{O} \right]_m$
P(CPM-SA)	$\left[\text{C}(=\text{O})-\text{C}_6\text{H}_4-(\text{CH}_2)_1-\text{C}_6\text{H}_4-\text{C}(=\text{O})-\text{O} \right]_n \left[\text{C}(=\text{O})-(\text{CH}_2)_8-\text{C}(=\text{O})-\text{O} \right]_m$
P(CPH-SA)	$\left[\text{C}(=\text{O})-\text{C}_6\text{H}_4-(\text{CH}_2)_6-\text{C}_6\text{H}_4-\text{C}(=\text{O})-\text{O} \right]_n \left[\text{C}(=\text{O})-(\text{CH}_2)_8-\text{C}(=\text{O})-\text{O} \right]_m$

Different polyanhydrides studied for various applications as localized drug carriers

Disease	Polymer	Delivery system	Drugs
Cancer	Ricinoleic acid based polyanhydride	Matrix	Methotrexate
	P(RA-SA)	Matrix	Cisplatin
	P(CPP-SA)	Implant	Bromodeoxyuridine
			N-(phosphonacetyl)-aspartic acid
	P(CPP-SA)	Implant	5-fluorouracil
	P(FAD-SA)	Implant	Taxol
	P(CPP-SA)	Implant	Camptothecin
	P(RA-SA)	Injectable Paste	Paclitaxel
Head and neck cancer	P(FAD-SA)	Matrix	Cisplatin, 5-FU, methotrexate, paclitaxel
Osteomyelitis	P(OA/LAD-SA)	Matrix	Gentamicin
Local anesthesia	P(FAD-SA)	Matrix	Bupivacaine HCl
	P(CPP-SA)	Implant	Dibucaine, bupivacaine
Local infection	P(DDDA-TA), P(BA-PA)	Matrix	Ciprofloxacin hydrochloride
	P(FAD-SA)	Implant	Cefazolin sodium
Restenosis	P(EAD-SA)	Implants	Heparin
Gene delivery	Photocrosslinked polyanhydride	Matrix	DNA
Glaucoma	P(CPP-SA)	Implant	Etoposide
Inflammatory bowel disease	Poly(anhydride-esters)	Microspheres	Aminosalicylates
Inflammation	PLA-PSA-PLA	Microspheres	Triamcinolone
Hormone therapy	P(FAD-SA)	Microsphere	GnRHa
Alzheimer's disease	SA copolymer	Microspheres	Bethanechol

Commercial Therapeutics

Proteins, DNA and other biomolecules are highly evolved and specific, so they make excellent drugs

- **Insulin**
 - Helps to regulate blood glucose levels for people with diabetes.
- **Interferon- α (Intron A, Roferon)**
 - Used for the treatment of chronic hepatitis C in adults.

Biomolecules such as proteins rapidly degrade in the body by natural mechanisms

This means that in order to have a sustained affect – the patient must endure many injections

